

Quick Check

Lesson 5: Feedback Loops: How the Body Keeps Water and Blood Glucose Steady

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 3.3: Excretion and Homeostasis in Animals | Lesson 5

Wanjiku runs 10 km on a hot afternoon in Kericho. She sweats a lot and drinks very little water, so her blood becomes more concentrated. What will her kidneys most likely produce?

- A** A small volume of concentrated urine, to save body water
- B** Urine with a lot of glucose, because running uses sugar
- C** A large volume of dilute urine, to remove the extra water
- D** The same amount of urine as normal, because the kidney does not respond to the body's water level

In osmoregulation, more ADH is released when the blood is too concentrated. Why does more ADH lead to less urine?

- A** ADH makes the bladder store water so it is not passed out
- B** ADH makes the glomerulus filter less blood, so no water enters the nephron
- C** ADH is a waste product that carries water out of the body
- D** ADH makes the collecting ducts more permeable, so more water is reabsorbed into the blood

After eating a big plate of ugali, a person's blood glucose rises above its set point. Which statement correctly describes the negative feedback response?

- A** The kidney alone decides to filter out the extra glucose, and no hormone is needed
- B** Insulin removes sugar from the body by carrying it out in the urine
- C** Beta cells in the pancreas release insulin, so liver and muscle take up glucose and store it as glycogen, and blood glucose falls back
- D** Alpha cells release glucagon, which makes the liver release more glucose into the blood

A runner's blood glucose falls low after a long run. Which hormone is released, and which organ responds to bring blood glucose back up?

- A** Glucagon from the beta cells; the muscles change glucose into urine
- B** ADH from the pituitary; the collecting duct releases glucose into the blood
- C** Glucagon from the alpha cells; the liver changes glycogen into glucose
- D** Insulin from the alpha cells; the kidney changes glucose into glycogen

A patient in Kisumu passes up to 10 litres of very dilute urine each day and is always thirsty. Tests show normal blood glucose and no glucose in the urine. Which step in the body's control system has most likely failed?

- A** Beta cells release too little insulin
- B** The proximal convoluted tubule cannot reabsorb glucose
- C** Alpha cells release too much glucagon
- D** ADH is missing, or the kidney does not respond to it, so too little water is reabsorbed

Normally all the glucose that is filtered into a nephron is reabsorbed in the proximal convoluted tubule, so urine has no glucose. Why does glucose appear in the urine of a person with diabetes mellitus?

- A** The kidney makes glucose from water when ADH is low
- B** Too little insulin makes blood glucose very high, so more glucose is filtered than the tubule can reabsorb
- C** Too much ADH stops the collecting duct from reabsorbing glucose
- D** Glucagon destroys the glucose before it reaches the nephron

A goat on a hot afternoon produces concentrated urine, but a freshwater fish produces large amounts of dilute urine. Which explanation is best?

- A** Both animals produce the same urine, but the fish looks like it makes more because it lives in water
- B** The goat has a high blood sugar level, which makes its urine concentrated
- C** The goat conserves water, with high ADH and more water reabsorbed, while the fish gains water by osmosis and must excrete the extra as dilute urine
- D** The fish has no kidneys, so its urine is just water from its gills

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).